

FOST TRAINING RULE OF THE ROAD TEST MAR 16

This test is based on: The International Regulations for Preventing Collisions at Sea (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(F) Vol I and the Mariners Handbook (NP100).

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the OOW training records for inspection by FOST navigation staff. The exam should take no longer than 1 hr.

In this test, any change in the status of the ship or of the environment will be indicated in the script. Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90% (45 out of 50)

You are an OOW in HMS DARING sailing from Portsmouth in order to conduct a work up period in the Eastern Exercise Areas and short notice Air Defence Demonstration in the North Sea in preparation for FOST next month.

It is the end of March; the weather is fine with visibility in excess of 10 NM although as always with the typical warm sector conditions Fog is predicted. The wind is South-Westerly 10-12 knots Sunrise was at 0615 and Sunset is at 1852.

1. Having successfully sailed from Portsmouth and passed OSB, you are now in the main channel outbound when you sight a small power-driven vessel at R20, range 1NM. You take an initial bearing and observe that you are 10 degrees on the starboard bow of the vessel. The DME0 asks if the Rules apply to this vessel. What is your answer and to whom do they apply? (R. 1a) (2)

2. The vessel acts appropriately and you pass clear. The YO asks if Portsmouth counts as a Narrow channel, how should you proceed along the course of a narrow channel or fairway? (R. 9a) (1)

3. On VHF you hear Solent Port Authority talking to the Pilot of a ship who says that ship is constrained by her draught.

a. What is this vessel required to display to indicate her special condition? (R.28) (1)

b. What lights would this vessel display if under way and making way, length 125 metres? (R.23 and R.28) (2)

c. What sound signal would you expect the accompanying pilot vessel engaged in pilotage duty to sound? (R.35k)

4. Having passed by NAB Tower, the NO hands you charge of the Ship while SSD fall out. He instructs you to proceed south along the Navtrack to your RV in order to embark your Flight.

a. When engaged in the launching or recovery of aircraft what status can your vessel claim? (R.3g) (1)

b. If you claim this status by day what shapes should you display? (R.27) (1)

5. With your flight safely embarked you head towards your assigned Exercise Area South of the Isle of Wight the visibility begins to reduce. Apart from at night, when should your navigation lights be displayed? (R.20c)

(2)

6. Visibility continues to reduce and your OPS Officer notifies FOST OPS that your Flying serial will be postponed. The CO decides to patrol an area awaiting an improvement in visibility. Whilst proceeding at slow speed you hear a fog signal consisting of one prolonged blast followed by two short blasts at approximately Red 50.

a. Name 3 types of vessel this could be. (R.35) (3)

b. If you stop your vessel in the water what fog signal shall you sound and at what interval? (R.35)

(2)

7. You detect using the bridge radar the presence of a contact at Green 30 range 3.5 miles, and determine that a close quarters situation is developing and/or risk of collision exists. What action shall you avoid taking? (R.19 d) (2)

Visibility opens up as quickly as it reduced, and you sight a merchant vessel corresponding with the previous radar contact. You ascertain that you are on a steady bearing with merchant vessel, and are required to take action to avoid collision.

8. You take avoiding action in good time but how large an alteration should be made to avoid collision, what do you say iaw the rules and what should be avoided (R. 8b) (1)

9. The CO returning to the Bridge comments that a Dutch Submarine is operating in the FOST Exercise areas during the week. The CO asks the YO whether or not the submarine must comply with the provisions of the Rules with regard to the display of navigation lights. What do the Rules state? (R. 1e) (2)

10. If you were to sight navigation lights of a vessel apparently clustered together and additionally showing an amber flashing light. What is this likely to be? (NP100 10th edition 2015) (1)

11. What distance is recommended to pass clear astern of this vessel? (NP100 10th edition 2015) (1)

Later on that afternoon the Ship is required to conduct a boat transfer for essential stores in Solent.

12. As you proceed along the main channel you need to overtake a small slow-moving coaster ahead which must take action to allow such passing. What sound signal should you sound to indicate you wish to overtake her on her starboard side? (R.34c) (2)

13. The CO wishing to test the NO asks whether a fast sailing vessel were to approach from an angle of greater than 22.5 degrees abaft the beam would we, as a power driven vessel, be the give way vessel? (R. 13b & R. 18) (1)

14. Would the answer to Q13 be different if a cone, apex down, was displayed on board the sailing vessel? (R. 13b & R. 18) (2)

The transfer of stores is completed and the Ship now tasked to proceed to the North Sea for a short notice Air Defence Demonstration.

15. It is the Dogs and you are back on watch whilst the Ship transit towards the Dover TSS. You sight two MCMV's ahead conducting exercises, they are displaying the shapes for conducting mine clearance operations and ahead is a helicopter in the dip.

a. How far should you pass clear of the helicopter in the dip? (ATP1F) (1)

b. What action should the MCMV's take when approaching a helicopter in the dip? (ATP1F) (1)

16. You commence the transit through Dover TSS.
- a. How do the rules say a vessel using the separation scheme shall operate? (R.10b i, ii, iii) (3)
 - b. When operating within a TSS, a vessel restricted in her ability to manoeuvre, engaged in an operation for the maintenance of safety of navigation, is exempted from what, and to what extent? (R.10k) (2)
17. Your SOA is high to meet your RV for the start of the Demonstration, what factors should be considered when determining safe speed in any state of visibility? (R.6) (6)

Passing Calais you see a ferry is inbound to Dover and currently at G30, range 4 nm. You are approximately 40 degrees on her PORT bow.

18. In determining if ROC exists what considerations should be taken into account? (R.7) (2)

You ascertain that a risk of collision does exist.

19. a. Who is the give-way vessel and why? (R.15) (2)
- b. What action should each vessel take? (R.15, R.16 & R.17) (3)

20. You are having a discussion with the YO on the type of distress signals. List 6 of the distress signals you could expect to receive. (Annex IV) (3)

ANSWERS

1. Having successfully sailed from Portsmouth and passed OSB, you are now in the main channel outbound when you sight a small power-driven vessel at R20, range 1NM. You take an initial bearing and observe that you are 10 degrees on the starboard bow of the vessel. The DMEO asks if the Rules apply to this vessel. What is your answer and to whom do they apply? **(R. 1a)** (2)

These Rules apply to all vessels upon the high seas and in all waters connected therewith navigable by seagoing vessels.

2. The vessel acts appropriately and you pass clear. The YO asks if Portsmouth counts as a Narrow channel, how should you proceed along the course of a narrow channel or fairway? (R. 9a) (1)

A vessel proceeding along the course of a narrow channel or fairway shall keep as near to the outer limit of the channel or fairway which lies on her starboard side as is safe and practicable.

3. On VHF you hear Solent Port Authority talking to the Pilot of a ship who says that ship is constrained by her draught.

a. What is this vessel required to display to indicate her special condition? (R.28) (1)

Exhibited where best seen a black cylinder, (diameter of at least 0.6 metre and a height of twice its diameter).

b. What lights would this vessel display if under way and making way, length 125 metres? (R.23 and R.28) (2)

Where best seen - three all round red lights in a vertical line plus fwd and aft masthead lights, sidelights and a stern light.

c. What sound signal would you expect the accompanying pilot vessel engaged in pilotage duty to sound? (R.35k)

A pilot vessel when engaged on pilotage duty may in addition to the signal prescribed in paragraphs (a), (b) or (g) of this rule sound an identity signal consisting of four short blasts.

4. Having passed by NAB Tower, the NO hands you charge of the Ship while SSD fall out. He instructs you to proceed south along the Navtrack to your RV in order to embark your Flight.

a. When engaged in the launching or recovery of aircraft what status can your vessel claim? (R.3g) (1)

Restricted in Ability to Manoeuvre.

b. If you claim this status by day what shapes should you display? (R.27) (1)

Three shapes in a vertical line where they can best be seen. The highest and lowest of these shapes shall be balls and the middle one a diamond;

5. With your flight safely embarked you head towards your assigned Exercise Area South of the Isle of Wight the visibility begins to reduce. Apart from at night, when should your navigation lights be displayed? (R.20c) (2)

The lights prescribed by these Rules shall, if carried, also be exhibited from sunrise to sunset in restricted visibility and may be exhibited in all other circumstances when it is deemed necessary.

6. Visibility continues to reduce and your OPS Officer notifies FOST OPS that your Flying serial will be postponed. The CO decides to patrol an area awaiting an improvement in visibility. Whilst proceeding at slow speed you hear a fog signal consisting of one prolonged blast followed by two short blasts at approximately Red 50.

a. Name 3 types of vessel this could be. (R.35) (3)

Vessel Not Under Command, Vessel Restricted in her ability to manoeuvre, vessel constrained by her draft, a sailing vessel, a vessel engaged in fishing or a vessel engaged in towing.

b. If you stop your vessel in the water what fog signal shall you sound and at what interval? (R.35) (2)

A power-driven vessel underway but stopped and making no way through the water shall sound at intervals of not more than 2 minutes two prolonged blasts in succession with an interval of about 2 seconds between them.

7. You detect using the bridge radar the presence of a contact at Green 30 range 3.5 miles, and determine that a close quarters situation is developing and/or risk of collision exists. What action shall you avoid taking? (R.19 d) (2)

A vessel which detects by radar alone the presence of another vessel shall determine if a close quarters situation is developing and/or risk of collision exists. If so, she shall take avoiding action in ample time, provided that when such action consists of an alteration of course, so far as possible the following shall be avoided:

(i) an alteration of course to port for a vessel forward of the beam, other than for a vessel being overtaken;

(ii) an alteration of course towards a vessel abeam or abaft the beam.

Visibility opens up as quickly as it reduced, and you sight a merchant vessel corresponding with the previous radar contact. You ascertain that you are on a steady bearing with merchant vessel, and are required to take action to avoid collision.

8. You take avoiding action in good time but how large an alteration should be made to avoid collision, what do you say iaw the rules and what should be avoided (R. 8b) (1)

It shall be large enough to be readily apparent to another vessel observing visually or by radar; a succession of small alterations of course and/or speed should be avoided.

9. The CO returning to the Bridge comments that a Dutch Submarine is operating in the FOST Exercise areas during the week. The CO asks the YO whether or not the submarine must comply with the provisions of the Rules with regard to the display of navigation lights. What do the Rules state? (R. 1e) (2)

Whenever the Government concerned shall have determined that a vessel of any special construction or purpose cannot comply with the provisions of any of these Rules with respect to the number, position, range or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signalling appliances, such vessel shall comply with such other provisions in regard to the number, position, range or arc of visibility of lights or shapes, as well as to the disposition and characteristics of sound-signalling appliances, as her Government shall have determined to be the closest possible compliance with these Rules in respect of that vessel.

10. If you were to sight navigation lights of a vessel apparently clustered together and additionally showing an amber flashing light. What is this likely to be? (NP100 10th edition 2015) (1)

All submarines are fitted with a flashing light to indicate to an approaching vessel the need for caution. ..typically yellow, orange or amber flashing at 70-180 flashes per minute or flashing (3) at a short interval.

11. What distance is recommended to pass clear astern of this vessel? (NP100 10th edition 2015) (1)

Submarines occasionally tow sonar equipment and other vessels are recommended to remain 1500 metres clear when crossing astern of a surfaced submarine.

Later on that afternoon the Ship is required to conduct a boat transfer for essential stores in Solent.

12. As you proceed along the main channel you need to overtake a small slow-moving coaster ahead which must take action to allow such passing. What sound signal should you sound to indicate you wish to overtake her on her starboard side? (R.34c) (2)

A vessel intending to overtake another shall in compliance with Rule 9(e)(i) indicate her intention by the following signals on her whistle:

- two prolonged blasts followed by one short blast to mean “I intend to overtake you on your starboard side”

13. The CO wishing to test the NO asks whether a fast sailing vessel were to approach from an angle of greater than 22.5 degrees abaft the beam would we, as a power driven vessel, be the give way vessel? (R. 13b & R. 18) (1)

No. The Sailing Vessel, as the Over taking Vessel, would be Give-Way. Rule 18 states ‘Except where Rules 9, 10 and 13 otherwise require’ negating 18aiv – A Power driven vessel shall keep out of the way of a sailing vessel.

14. Would the answer to Q13 be different if a cone, apex down, was displayed on board the sailing vessel? (R. 13b & R. 18) (2)

No. Even if the other vessel was classed as a power driven vessel then she would still be the overtaking vessel and therefore give way.

The transfer of stores is completed and the Ship now tasked to proceed to the North Sea for a short notice Air Defence Demonstration.

15. It is the Dogs and you are back on watch whilst the Ship transit towards the Dover TSS. You sight two MCMV's ahead conducting exercises, they are displaying the shapes for conducting mine clearance operations and ahead is a helicopter in the dip.

- a. How far should you pass clear of the helicopter in the dip? (ATP1F) (1)

A helicopter in the dip is considered as a ship not under command and no vessels should approach within 500 yards.

- b. What action should the MCMV's take when approaching a helicopter in the dip? (ATP1F) (1)

The mine hunter would be the give way vessel and should keep out of the way.

16. You commence the transit through Dover TSS.

- a. How do the rules say a vessel using the separation scheme shall operate? (R.10b i, ii, iii) (3)

i Proceed in the appropriate traffic lane in the general direction of traffic flow for that lane

ii so far as practicable keep clear of the traffic separation line of separation zone

iii Normally join or leave a traffic lane at the termination of the lane, but when joining or leaving from either side shall do so at as small an angle to the general direction of traffic flow as practicable.

- b. When operating within a TSS, a vessel restricted in her ability to manoeuvre, engaged in an operation for the maintenance of safety of navigation, is exempted from what, and to what extent? (R.10k) (2)

A vessel restricted in her ability to manoeuvre when engaged in an operation for the maintenance of safety of navigation in a traffic separation scheme is exempted from complying with this rule to the extent necessary to carry out the operation.

17. Your SOA is high to meet your RV for the start of the Demonstration, what factors should be considered when determining safe speed in any state of visibility? (R.6) (6)

- (i) the state of visibility;**
- (ii) the traffic density including concentrations of fishing vessels or any other vessels;**
- (iii) the manoeuvrability of the vessel with special reference to stopping distance and turning ability in the prevailing conditions;**
- (iv) at night the presence of background lights such as from shore lights or the backscatter of her own lights;**
- (v) the state of wind, sea and current, and the proximity of navigational hazards;**
- (vi) the draught in relation to the available depth of water.**

Passing Calais you see a ferry is inbound to Dover and currently at G30, range 4 nm. You are approximately 40 degrees on her PORT bow.

18. In determining if ROC exists what considerations should be taken into account? (R.7) (2)

In determining if risk of collision exists the following considerations shall be among those taken into account:

(i) such risk shall be deemed to exist if the compass bearing of an approaching vessel does not appreciably change;

(ii) such risk may sometimes exist even when an appreciable bearing change is evident, particularly when approaching a very large vessel or a tow or when approaching a vessel at close range.

You ascertain that a risk of collision does exist.

19. a. Who is the give-way vessel and why? (R.15) (2)

You are, this is a crossing situation under R.15 and the ferry is on your starboard side.

c. What action should each vessel take? (R.15, R.16 & R.17) (3)

Ferry – keep her course and speed

HMS DARING – take early and substantial action to keep well clear and ... if the circumstances of the case admit, avoid crossing ahead of the other vessel.

20. You are having a discussion with the YO on the type of distress signals. List 6 of the distress signals you could expect to receive. (Annex IV)

A gun or other explosive signal fired at intervals of about a minute

A continuous sounding with any fog signal

A signal made by radiotelephony or by any other signalling method consisting of the group ... --- ... (SOS) in Morse code

A signal sent by radiotelephony consisting of the spoken word mayday

The international code signal of distress indicated by NC

The radiotelegraph alarm signal

The radiotelephone alarm signal

Signals transmitted by emergency positioning-indicating radio beacons

Approved signal transmitted by radio communications systems, including survival craft radar transponders